RETINAX - FRIEND SETUP GUIDE

WHAT THIS PROJECT DOES

RetinaX classifies fundus images into five diabetic-retinopathy severity classes, creates Grad-CAM explanations, and can localize one lesion type with Attention U-Net.

1. INSTALL ON THE GAMING PC

```
git clone <YOUR-REPOSITORY-URL>
cd retinax
python -m venv .venv
.venv\Scripts\activate        (Windows)
source .venv/bin/activate      (macOS/Linux)
pip install -r requirements.txt
pip install -e .
```

2. CHECK THE NVIDIA GPU

```
python -c "import torch; print(torch.cuda.is_available()); print(torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'CPU')"
```

The first result should be True and the second should show the NVIDIA GPU name.

3. DOWNLOAD THE CLASSIFICATION DATASET

Download from Kaggle:
https://www.kaggle.com/datasets/sovitrath/diabetic-retinopathy-224x224-gaussian-filtered

Create a free Kaggle account if needed. The folder structure must be:

```
classification_dataset/
  Mild/
  Moderate/
  No_DR/
  Proliferate_DR/
  Severe/
```

Verify it:
```
python -m retinax.cli inspect-data --data-root "D:\datasets\gaussian_filtered_images"
```

4. TRAIN THE CLASSIFICATION MODEL

```
python -m retinax.cli train --data-root "D:\datasets\gaussian_filtered_images" --model baseline --epochs 15 --batch-size 32 --checkpoint "checkpoints\retinax_baseline.pt"
```

This creates checkpoints/retinax_baseline.pt for severity prediction, confidence scores, and Grad-CAM.

5. DOWNLOAD THE SEGMENTATION DATASET

Download A. Segmentation.zip from the official IDRiD IEEE DataPort page:
https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid

A free IEEE account is enough; IEEE membership is not required.

After extraction:
A. Segmentation/
  1. Original Images/
  2. All Segmentation Groundtruths/

Keep the license files and cite IDRiD in reports.

6. TRAIN LESION LOCALIZATION

```
python -m retinax.cli train-segmentation --images "D:\datasets\A. Segmentation\1. Original Images" --masks "D:\datasets\A. Segmentation\2. All Segmentation Groundtruths" --lesion MA --epochs 20 --batch-size 4 --checkpoint "checkpoints\retinax_unet_ma.pt"
```

Lesion codes:
MA = microaneurysms
HE = hemorrhages
EX = hard exudates
SE = soft exudates

7. RUN THE UI

```
streamlit run app.py
```

Open http://localhost:8501.

8. USE EVERY UI CONTROL

Classification model: choose the model that was trained. Use Baseline CNN for the command above.
Classification checkpoint: checkpoints/retinax_baseline.pt
Segmentation checkpoint: checkpoints/retinax_unet_ma.pt
Upload: choose a JPG or PNG fundus image.
Analyse image: view severity, probabilities, Grad-CAM, and lesion mask.
First-time setup commands: expand the sidebar section to see the commands again.

9. WHAT THE OUTPUTS MEAN

Severity: model estimate among No_DR, Mild, Moderate, Severe, and Proliferate_DR.
Confidence: probability distribution, not proof of correctness.
Grad-CAM: regions influencing the classifier, not proof of a real lesion.
Lesion localization: mask for the lesion type used during segmentation training.

10. WHAT TO PUSH TO GIT

Push source code and documentation. Do not push datasets or large checkpoints.
Your friend can train checkpoints on her gaming PC. Alternatively, share .pt files separately.

TROUBLESHOOTING

Checkpoint not found: train the model first and confirm the checkpoints folder.
No image/mask pairs: use the original IDRiD folder paths and preserve filenames.
CUDA is False: PyTorch is not seeing the NVIDIA GPU.
Untrained preview: no valid trained checkpoint is loaded.

IMPORTANT: RetinaX is a research prototype, not a medical diagnostic device.